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Agile Development Process and DevOps

Lab Assessment : 3

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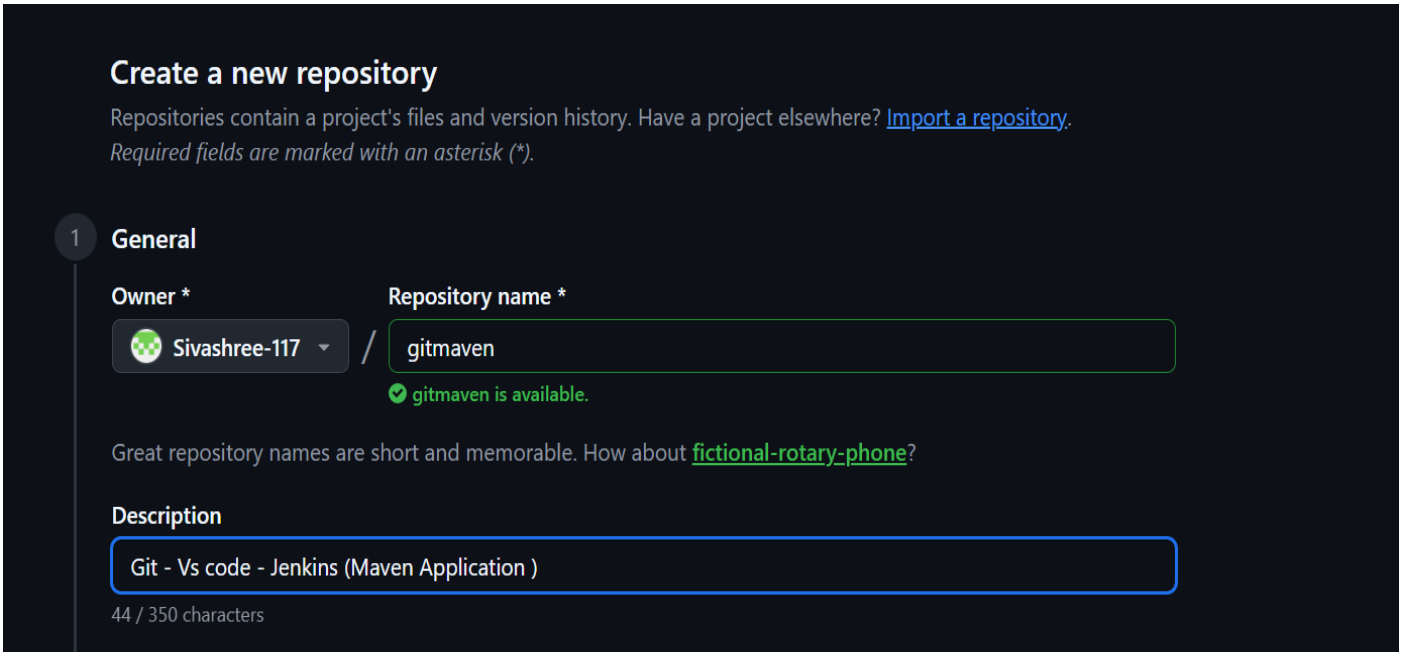
- **Reg No :** 23MIS0117
- **Course Title :** Agile Development Process
- **Course code :** ISWE406P
- **Faculty :** Dr. Sweta Bhattacharya
- **Slot :** L49 + L50

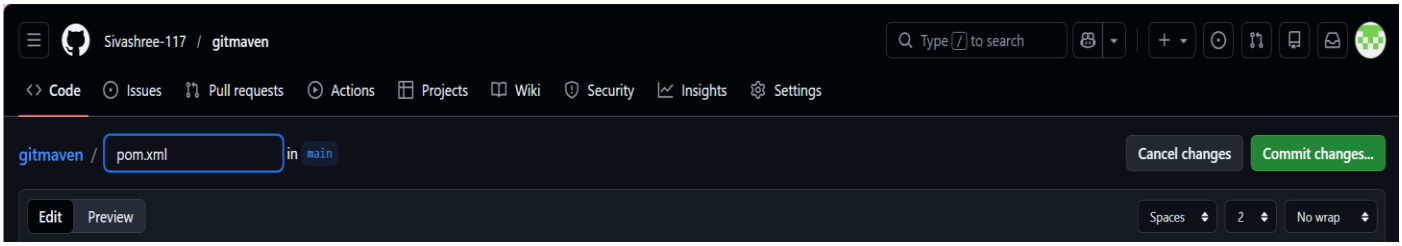
1. Create a Maven Project in Git

Create a new Git repository

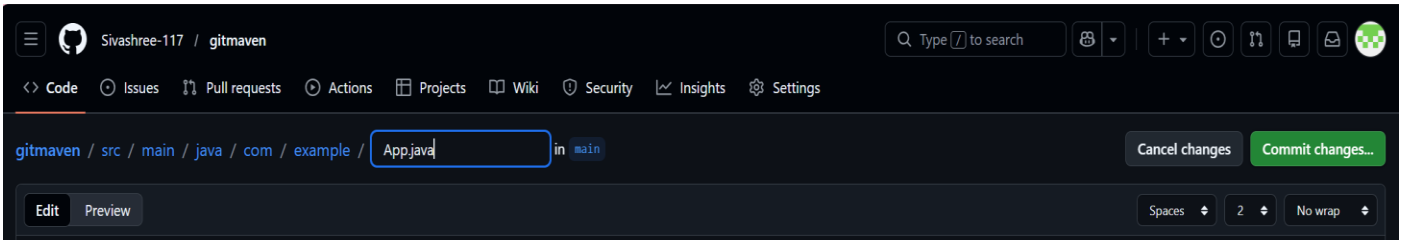


Create a simple Maven project structure manually, including the following files and folders

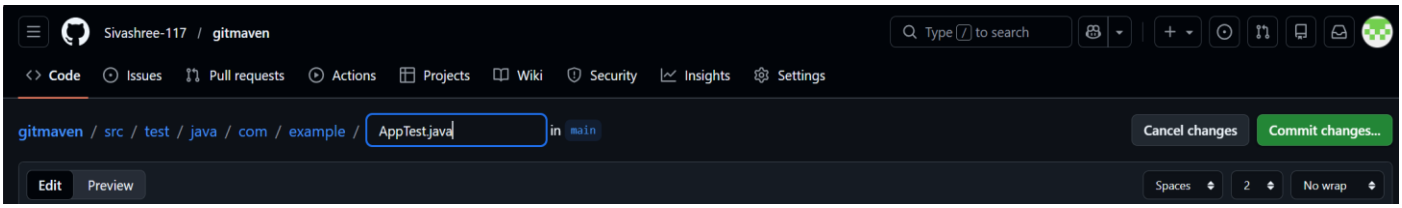
pom.xml



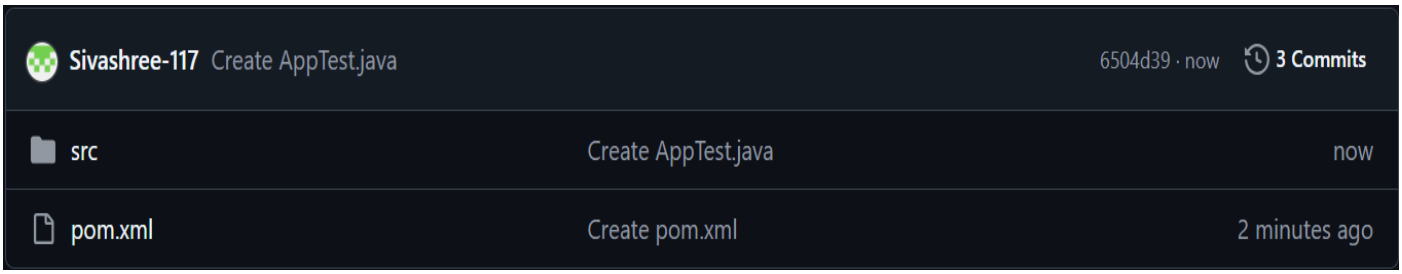
src/main/java/com/example/App java

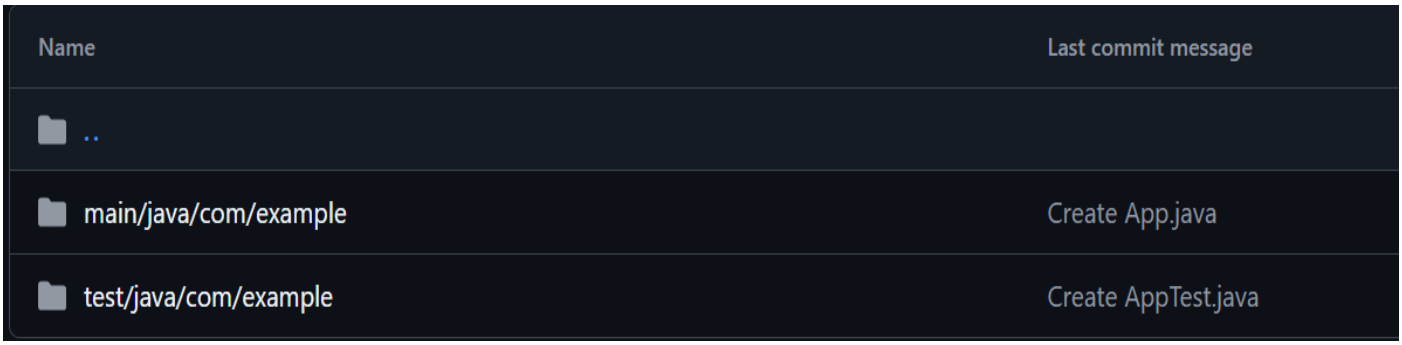


src/test/java/com/example/AppTest.java



File structure :





2. Implement the Java Program

Write a simple Java program in App.java under your newly created Git repository for any one of the following scenarios

gitmaven / pom.xml

Sivashree-117 Update pom.xml

Code Blame 31 lines (27 loc) · 956 Bytes

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
         http://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <groupId>com.example</groupId>
    <artifactId>student-grade</artifactId>
    <version>1.0-SNAPSHOT</version>

    <dependencies>
        <!-- JUnit 5 -->
        <dependency>
            <groupId>org.junit.jupiter</groupId>
            <artifactId>junit-jupiter</artifactId>
            <version>5.9.3</version>
            <scope>test</scope>
        </dependency>
    </dependencies>

    <build>
        <plugins>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-surefire-plugin</artifactId>
                <version>3.0.0</version>
            </plugin>
        </plugins>
    </build>
</project>
```

Student Grade Calculator

gitmaven / src / main / java / com / example / App.java in main

Edit Preview

```java
package com.example;

public class App {

    public static String calculateGrade(int marks) {

        if (marks >= 90) return "A";
        else if (marks >= 75) return "B";
        else if (marks >= 60) return "C";
        else if (marks >= 50) return "D";
        else return "F";
    }

    public static void main(String[] args) {
        int marks = 85;
        String grade = calculateGrade(marks);
        System.out.println("Marks: " + marks);
        System.out.println("Grade: " + grade);
    }
}
```

Write a corresponding unit test in AppTest.java

gitmaven / src / test / java / com / example / AppTest.java in main

Edit Preview

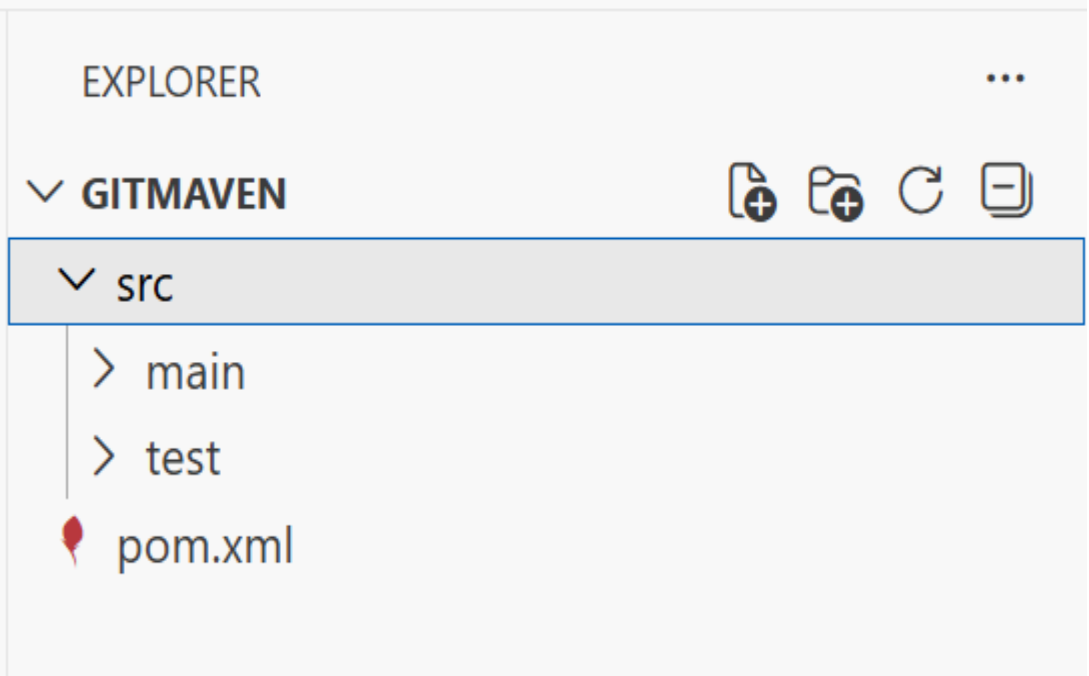
```java
package com.example;

import static org.junit.jupiter.api.Assertions.assertEquals;
import org.junit.jupiter.api.Test;

public class AppTest {

    @Test
    void testGradeA() {
        assertEquals("A", App.calculateGrade(95));
    }

    @Test
    void testGradeB() {
        assertEquals("B", App.calculateGrade(80));
    }

    @Test
    void testGradeFail() {
        assertEquals("F", App.calculateGrade(30));
    }
}
```

3. Push and Clone the Repository

Clone the repository using Visual Studio Code.

```
PS C:\Users\sivas\OneDrive\Desktop\git\gitmaven> git clone https://github.com/Sivashree-117/gitmaven.git
Cloning into 'gitmaven'...
remote: Enumerating objects: 36, done.
remote: Counting objects: 100% (36/36), done.
remote: Compressing objects: 100% (17/17), done.
remote: Total 36 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (36/36), 7.01 KiB | 1.40 MiB/s, done.
PS C:\Users\sivas\OneDrive\Desktop\git\gitmaven>
```

Verify the project structure.



4. Build the Maven Project

Use Maven commands in Visual Studio Code terminal to:

Compile the project

```
PS C:\Users\sivas\OneDrive\Desktop\Git\gitmaven> mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] ---------------------< com.example:student-grade >----------------------
[INFO] Building student-grade 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ student-grade ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\sivas\OneDrive\Desktop\Git\gitmaven\src\main\resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ student-grade ---
[INFO] Recompiling the module because of added or removed source files.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\classes
[WARNING] bootstrap class path is not set in conjunction with -source 8
  not setting the bootstrap class path may lead to class files that cannot run on JDK 8
    --release 8 is recommended instead of -source 8 -target 1.8 because it sets the bootstrap class path automat
ically
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.032 s
[INFO] Finished at: 2026-03-04T17:32:15+05:30
[INFO] ------------------------------------------------------------------------
```

Run tests

```
PS C:\Users\sivas\OneDrive\Desktop\Git\gitmaven> mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] ---------------------< com.example:student-grade >----------------------
[INFO] Building student-grade 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-surefire-plugin/3.
0.0/maven-surefire-plugin-3.0.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-surefire-plugin/3.0
.0/maven-surefire-plugin-3.0.0.pom (5.9 kB at 10 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire/3.0.0/surefire
-3.0.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire/3.0.0/surefire-
3.0.0.pom (22 kB at 396 kB/s)
```

```
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.066 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  10.695 s
[INFO] Finished at: 2026-03-04T17:32:49+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\sivas\OneDrive\Desktop\Git\gitmaven>
```

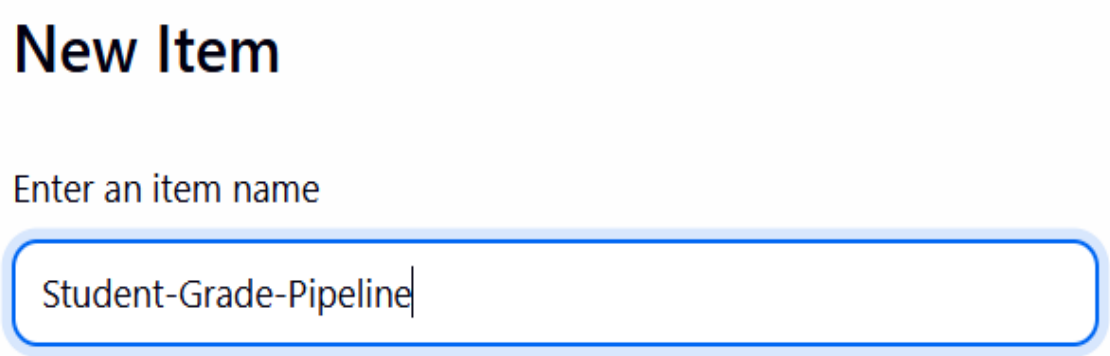
Build the project (mvn clean install)

```
PS C:\Users\sivas\OneDrive\Desktop\Git\gitmaven> mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] ---------------------< com.example:student-grade >----------------------
[INFO] Building student-grade 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ student-grade ---
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ student-grade ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\sivas\OneDrive\Desktop\Git\gitmaven\src\main\resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ student-grade ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\classes
[WARNING] bootstrap class path is not set in conjunction with -source 8
  not setting the bootstrap class path may lead to class files that cannot run on JDK 8
    --release 8 is recommended instead of -source 8 -target 1.8 because it sets the bootstrap class path automat
```

```
[INFO] Installing C:\Users\sivas\OneDrive\Desktop\Git\gitmaven\pom.xml to C:\Users\sivas\.m2\repository\com\exam
ple\student-grade\1.0-SNAPSHOT\student-grade-1.0-SNAPSHOT.pom
[INFO] Installing C:\Users\sivas\OneDrive\Desktop\Git\gitmaven\target\student-grade-1.0-SNAPSHOT.jar to C:\Users
\sivas\.m2\repository\com\example\student-grade\1.0-SNAPSHOT\student-grade-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  13.644 s
[INFO] Finished at: 2026-03-04T17:36:18+05:30
[INFO] ------------------------------------------------------------------------
```

5. Create and Run Jenkins Pipeline

Create a new Pipeline project in Jenkins.



New Item

Enter an item name

Student-Grade-Pipeline

Select an item type

Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

Maven project
Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.

Create a pipeline script

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

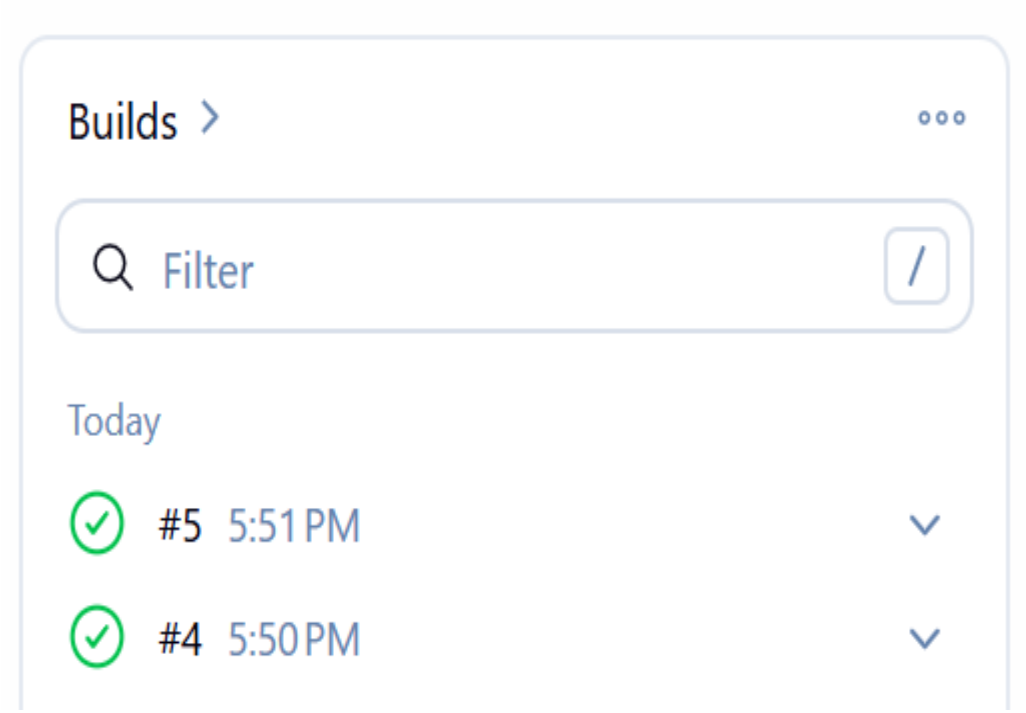
Script ?

try sample Pipeline...

```
pipeline {
    agent any

    tools {
        maven 'Maven'   // Make sure Maven is configured in Jenkins Global Tool Config
        jdk 'JDK'       // Make sure JDK is configured
    }

    stages {

        stage('Clone Repository') {
            steps {
                git 'https://github.com/yourusername/student-grade-maven.git'
            }
        }
```

Use Groovy Sandbox ?

Execute the pipeline and verify successful build completion




```
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.18/maven-resolver-util-1.9.18.jar (196 kB at 360 kB/s)
[INFO] Installing C:\ProgramData\Jenkins\.jenkins\workspace\Student-Grade-Pipeline\pom.xml to C:\WINDOWS\system32\config\systemprofile\.m2\repository\com\example\student-grade\1.0-SNAPSHOT\student-grade-1.0-SNAPSHOT.pom
[INFO] Installing C:\ProgramData\Jenkins\.jenkins\workspace\Student-Grade-Pipeline\target\student-grade-1.0-SNAPSHOT.jar to C:\WINDOWS\system32\config\systemprofile\.m2\repository\com\example\student-grade\1.0-SNAPSHOT\student-grade-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  30.635 s
[INFO] Finished at: 2026-03-04T17:51:32+05:30
[INFO] ------------------------------------------------------------------------
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Stage View

	Clone Repository	Compile	Test	Build
Average stage times: (full run time: ~50s)	2s	5s	7s	10s
#5 Mar 04 17:51 No Changes	2s	3s	6s	9s
#4 Mar 04 17:50 No Changes	2s	16s	23s	33s

OUTPUT :

```
C:\Windows\System32\cmd.e

Microsoft Windows [Version 10.0.26200.7922]
(c) Microsoft Corporation. All rights reserved.

C:\ProgramData\Jenkins\.jenkins\workspace\Student-Grade-Pipeline\target>java -cp student-grade-1.0-SNAPSHOT.jar com.example.App
Marks: 85
Grade: B

C:\ProgramData\Jenkins\.jenkins\workspace\Student-Grade-Pipeline\target>
```